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Data-Driven Optimization of Hospitality Operations:

A Case Study of Hotel SR Paark, Kavali

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Final Report for the Project on Business Data Management

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Declaration Statement

I am working on a Project Title "**Data-Driven Optimization of Hospitality Operations: A Case Study of Hotel SR Paark, Kavali**". I extend my appreciation to **Hotel SR Paark**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:

Name: Mannepalli Bala Praharsha

Date: 30-01-2026

1. EXECUTIVE SUMMARY AND TITLE

Project Title:

Data-Driven Optimization of Hospitality Operations: A Case Study of Hotel SR Paark, Kavali

Hotel SR Paark, Kavali, is a mid-scale hospitality service provider that provides room accommodation and banquet services. Although the hotel has maintained its operations in the market, the revenue has been fluctuating as a result of excessive seasonality in demand, poor weekday occupancy. Also, reliance on manual record-keeping has reduced the management capacity of implementing data-based pricing models and analyzing the revenue opportunities of add-on services systematically.

In this project, the primary data operations are analyzed based on the data gathered in the calendar year 2025, which encompasses room bookings, banquet activities, and add-on services sales. The measures of occupancy, revenue distribution, and demand variability were evaluated by descriptive statistical analysis. The time-series trend analysis was used to identify the demand patterns in seasons, the correlation analysis was used to establish the relationship between the room occupancy and the consumption of add-on services. These approaches have been chosen to directly deal with the inefficiencies of operations and pricing that the hotel experienced.

As can be seen in the analysis, the average room occupancy is low at approximately 34% of total capacity, which shows that it is underutilised. The highest revenue is observed in May (₹13,44,200), which is mainly caused by the demand related to weddings, and the first quarter has the lowest level of performance. There exists a very good positive relationship between occupancy and add-on services revenue that exists in peak seasons.

According to these results, it is proposed in the project to apply the dynamic pricing model, such as a discount of 15-20% during midweek on room services and a loyalty program of different levels and stages to ensure that potential clients can be retained. These actions should enhance the off-season income by around 12-15 percent. The hotel has, as a direct consequence, embarked on standardized digital record-keeping to aid in better forecasting as well as in decision making.

2. DETAILED EXPLANATION OF ANALYSIS PROCESS & METHOD

2.1. Data Collection and Preparation

Primary data for the study was collected directly from the physical booking registers of Hotel SR Paark for the period between January 2025 and December 2025. The information was captured on daily operational information that included the number of rooms in use, total revenue, use of supplemental beds, sales of cold drinks, type of stay, and comments. In order to have a systematic analysis, the data that had been recorded manually were digitized and exported into MS Excel. This organized information was the foundation of additional examination of occupancy rates and turnover.

2.2. Cleaning and Processing of Data

Process:

- Handwritten entries were computerized and made to a similar date format (MM-YYYY).
- There were missing values on add-on services, which were interpreted as zero.
- Redundant records and different notations were eliminated.
- The numerical variables, like the revenue and occupancy, were checked using logical accuracy.

This preprocessing was carried out so as to be able to have reliability and validity of the dataset when subjected to meaningful analysis.

2.3. Method 1: Analysis of Descriptive Statistics

About: Central tendencies and variability in occupancy, revenue and add-on services were summarized using descriptive statistics. Means and standard deviation measures were used to give a clue on average performance and variation of the hotel.

Justification: This approach gives a quantitative impression about the normal performance and variability of the hotel, directly focusing on the issue of fluctuating occupancy and revenues.

Importance: It gives a benchmark level of performance that is to be enhanced further and compared.

2.4. Method 2: Trend and Seasonality Analysis

About: The monthly occupancy and revenue trends were analyzed with the help of line and bar charts. An analysis of the trends allowed identifying the long-term trends and seasonal changes in demand.

Formula : $Seasonality\ Index = \left(\frac{Monthly\ Revenue}{Average\ Annual\ Monthly\ Revenue} \right)$

Justification: This is a method that averages fluctuations in the short run and reveals the long-term demand patterns using very little historical information.

Importance: It aids in active reservations and price policy when the demand is low in planning.

2.5. Method 3: Occupancy Rate Analysis.

About: This is a method to compute the percentage of the available room inventory that is sold during a given time. It is the main measure of asset utilization.

Formula: $Occupancy\ Rate\ (OR) = \left(\frac{Total\ Rooms\ Occupied}{Total\ Rooms\ Avaliable} \right) * 100$

Justification: This is needed to distinguish between the Slump and Peak periods of the issue statement. It gives a minimum demand for the rooms irrespective of the price.

Importance: It determines the days of the week (Tuesdays/Wednesdays) that the hotel is currently missing out on.

2.6. Method 4: Revenue per Available Room (RevPAR)

About: Unlike total revenue, it quantifies the amount of revenue being generated by each room (occupied or not).

Formula: $RevPAR\ = \left(\frac{Total\ Room\ Revenue}{Total\ Rooms\ Avaliable} \right)$

Justification: The technique aids in testing whether the hotel is over-discounting or the prices are the best in the peak seasons, such as in May.

Importance: It assists the management in knowing the financial well-being of the room inventory, in particular.

2.7. Method 5: Dynamic Pricing Model

About: A demand-based pricing adjustment was conceptualized by linking room pricing to occupancy levels. This was done to overcome the fixed pricing, which cannot realize increased revenue when demand is high.

Formula: $Adjusted\ Price = Base\ Price\ + (1 + \frac{Occupancy\ Rate}{100})$

Justification: Flat-rate pricing fails to capture revenue during peak demand periods.

Importance: Dynamic pricing enhances the maximization of revenue without necessarily raising the cost of operation.

2.8. Method 6: Revenue Contribution Analysis

About: Revenue has been studied by dividing the room revenue and add on service revenue. This assisted in determining the contribution of services that were given as supplements, like additional bed space and the sale of cold drinks.

Formula: $Total\ Revenue = Room\ Revenue + Add\ on\ Revenue$

Justification: Awareness of the auxiliary revenue streams can be used to establish unexploited profit potentials.

Importance: This approach will facilitate upselling with targeted measures to stabilize revenues in times of low occupancy.

2.9. Method 7: Correlation Analysis (Rooms vs. Add-ons)

About: This statistical device indicates the strength of the connection between the quantity of guests (Occupancy) and the sale of additional services (Cold drinks, extra beds).

Formula: $$r = \frac{n(\Sigma\, \mathrm{xy}) - (\Sigma\, \mathrm{x}) - (\Sigma\, \mathrm{y})}{\sqrt{[n\, \Sigma\, \mathrm{x}^2 - (\Sigma\, \mathrm{x})^2][n\, \Sigma\, \mathrm{y}^2 - (\Sigma\, \mathrm{y})^2]}}$$

Justification: It demonstrates whether or not the Add-on Services make sense based on the occupancy or whether they can be expanded as a separate source of revenue.

Importance: It substantiates the suggestion to cross-sell services at the front desk.

2.10. Tools Used

- MS Excel: Cleaning, analysis, forecasting, charts and visualization of data.
- Rational Connection to Problem Statement.

Each method directly addresses the identified problems of low occupancy, revenue instability, and ineffective pricing by providing data-driven insights and actionable strategies for optimization.

2.11 Reasoning of Method Chosen of Alternatives

Although there are various methods that can be used to analyse hospitality revenue, methods used in this research were based on the availability of data, interpretation and business applicability.

Demand modeling with regression was also to be taken; it was not useful as there was no control of pricing variation in the past. As room tariffs did not vary significantly with the demand situation, the regression coefficients would not give a stable and understandable estimate of the elasticity.

ARIMA time series forecasting models were also tested. The models take years of continuous historical data to make the forecast. Since only one year of daily data was at hand (2025), descriptive trend analysis and seasonality analysis were more useful with much less overfitting.

The reason why correlation analysis was used as opposed to multivariate causal models was to determine the directional and strength of correlation between occupancy and add-on services. The aim was knowledge of diagnosis and not forecasting.

Therefore, the best methods to tackle the operational inefficiencies, pricing rigidity, and volatility of revenues directly affecting the hotel were descriptive statistics, trend analysis, RevPAR and correlation.

3. RESULTS & FINDINGS

This section gives the results of the analysis of operational data during the period between January 2025 and December 2025 in Hotel SR Paark. Different charts and visual representations were applied to investigate room occupancy trends, revenue performance, and add-on services contribution. Such results are useful in determining the gaps in operations, demand trends in seasons, and revenue optimization.

3.1. Monthly Room Occupancy Trend

The monthly room occupancy trend indicates that there are conspicuous seasonal changes in demand throughout the year.

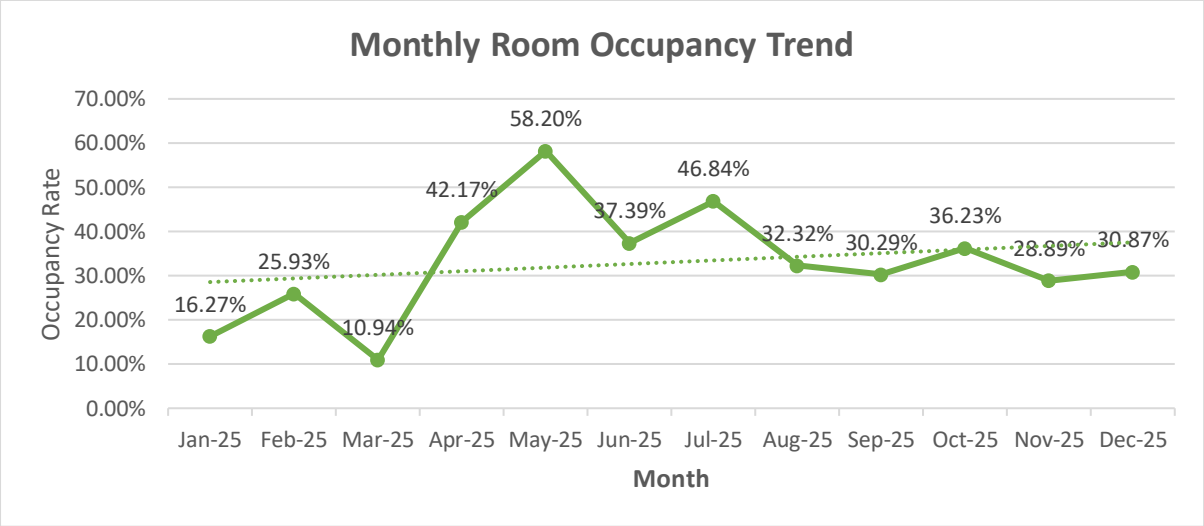

Figure 1: Monthly Room Occupancy Trend

- **Low Occupancy Period (January-March):**
  - The occupancy levels in the first quarter of the year are continually low. There is a decrease in occupancy with January as the highest of 16.27% to the lowest of 10.94% in March; the occupancy reflects the poor demand and underuse of room capacity. In this stage, almost 85-89 per cent of the total number of rooms is not occupied, and this leads to the loss of a lot of revenue and a lack of efficiency in the utilization of fixed operational resources.
- **Peak Demand Phase (April-May):**
  - A drastic increase in occupancy is noticed between April (42.17%) and the highest level of the year in May (58.20%). This is an increase over the levels in March, and it is during the time of summer travel and weddings. May achieves the greatest amount of room occupied throughout the year, which indicates good seasonal room demand as well as the ability of the hotel to receive bookings during prime times of the year.

- **Stabilization Phase (June-December):**
  - After hitting the peak in May, occupancy decreases, but stays between 28 and 46 per cent in the period between June and the end of December. Although demand is moderate in this season, the occupancy does not go all the way to the peak, meaning that seasonal demand is the only reason the performance survives without the proactive effort of creating demand or lowering prices.
- **Overall Trend Analysis:**
  - The dotted trend line shows that there is a slight positive change in the average occupancy throughout the year, which implies that there is a slow improvement in the performance. Nevertheless, the fact that there is a big difference in the number of peak and off-peak months shows that this hotel is highly reliant on the seasonal demand, but not the regular occupancy that is anticipated throughout the entire year.
- **Business Implication:**
  - The occupancy trend analysis proves that Hotel SR Paark has severe seasonality and uneven room occupancy with severe underutilization during the first months of the year and high dependence on the peak seasons to generate revenue. These results imply that weekday discount policies, price determined by occupancy, and specially selected promotions during low seasons are required to enhance consistency of utilization, minimize revenue variability, and optimize annual revenue potential.

3.2. Performance Efficiency – RevPAR Analysis

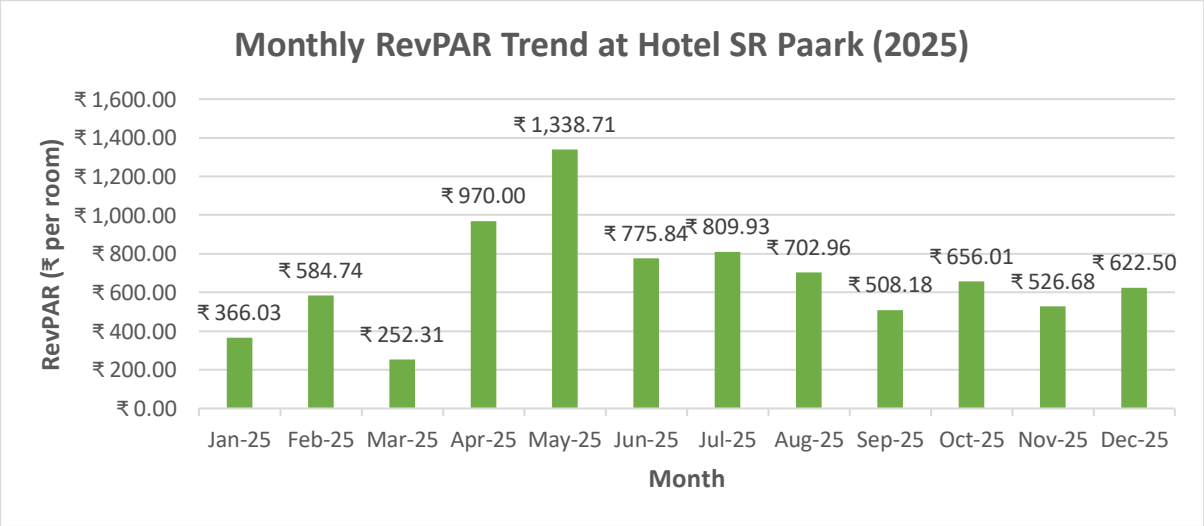

Figure 2: Monthly RevPAR Trend

- **Low RevPAR Period (January-March):**
  - The RevPAR values that are recorded in the first quarter of the year are low, which implies that there is low revenue generation on the available room inventory. RevPAR decreases gradually and has a minimum in March at ₹252.31, which is evidence of unfavorable demand situation and poor usage of the existing rooms. At this stage, though rooms are in use, the revenue collected per room is very low, which proves that there is high leakage of revenue and low financial efficiency of fixed assets.
- **Peak RevPAR Phase (April-May):**
  - It is possible to note a steep increase in the RevPAR in April (₹970.00), which peaks during the month of May (₹1,338.71). This is greater than the level in March, which is due to high seasonal demand during summer travel and the wedding season. The highest RevPAR in May shows the most effective monetization of room inventory as the high occupancy and the price remain constant to the maximum profit per available room. The company must maintain its market stance and concentrate on attracting new clients to its business.
- **Stabilization Phase (June-December):**
  - The company should retain its position in the market and focus on attracting new clients to the business. After peaking in May, RevPAR decreases and levels off within the range of ₹508 to ₹809 from June to December. Although these values indicate moderate performance, they are still well below the peaks, suggesting that revenue is constrained by constant pricing even though occupancy is reasonable. This stage suggests the reliance on the seasonal base demand, as opposed to the active, revenue maximization solutions.
- **overall RevPAR:**
  - The trend line shows that RevPAR is steadily improving throughout the year, though the broad difference between the peak and off-peak values depicts high revenue volatility. The lack of a smoother RevPAR trend indicates that they do not make use of pricing accommodation or revenue management strategies.
- **Business Implication:**
  - The RevPAR trend proves that Hotel SR Paark can only work efficiently during peak seasons and underperform during off-peak times. Dynamic pricing, weekday price reduction and demand-based promotion will be of great help in improving RevPAR in low-demand months and increase room level profitability.

3.3. Correlation Analysis: Room Occupancy vs. Add-on Services

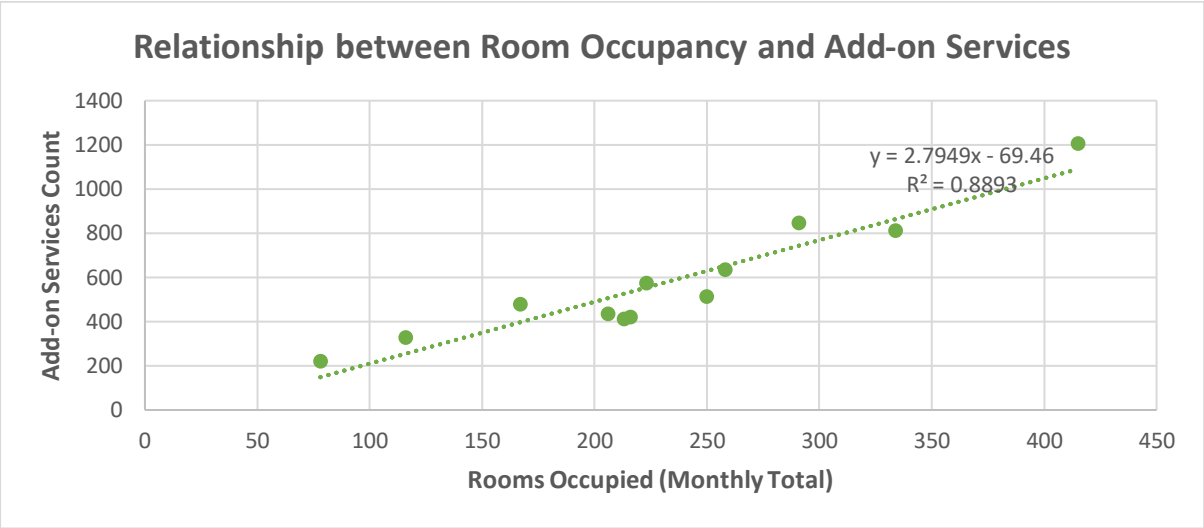

Figure 3: Relationship between Room Occupancy and Add-on Services

(This analysis examines how the usage of add-on services varies with changes in room occupancy.)

- **Character of Relationship between Occupancy and Add-on Usage:**
  - The scatter plot, with a fitted line trend, indicates the strong positive linear relationship of the monthly room occupancy with the use of additional services. The Pearson correlation coefficient of 0.943 suggests an extraordinarily high level of association, which shows that gains in the room occupancy are strictly accompanied by gains in the add-on consumption.
- **Strength of Dependence (Statistical Explanation):**
  - The coefficient of determination (R2 = 0.889) suggests that the variation in the use of add-on services can be attributed to the variation in room occupancy, about 89 per cent. This is a high explanatory power, which proves that add-on sales are highly subject to occupancy levels, and there is extremely little fluctuation that can be attributed to independent guest purchasing behavior. The upward-sloping trend line shows that the add-on sale is proportional to the occupied rooms.
- **Perspective and Business Intelligibility:**
  - This very high correlation indicates that add-on services at Hotel SR Paark are volume-based and not value-based. There is no significant difference in add-on purchases of guests who visit the hotel when the hotel is not busy with guests, versus when guests visit the hotel when the hotel is busy, indicating that there are no organized upselling or incentive programs.

- **Business Implication:**
  - The fact that occupancy and add-on revenue are strongly dependent creates volatility in the income and restricts diversification of revenue. The hotel will need to avoid depending on the occupancy-driven add-on sales and stabilize revenue by introducing the add-on bundles, add-on incentives, and front-desk proactive upselling. These measures will be able to raise the average revenue per guest, density of add-ons during low-demand times, and general revenue stability.

3.4. Distribution of Revenue per month

The occupancy pattern is almost similar to the monthly revenue trend, which depicts that the revenue is very dependent on the use of rooms.

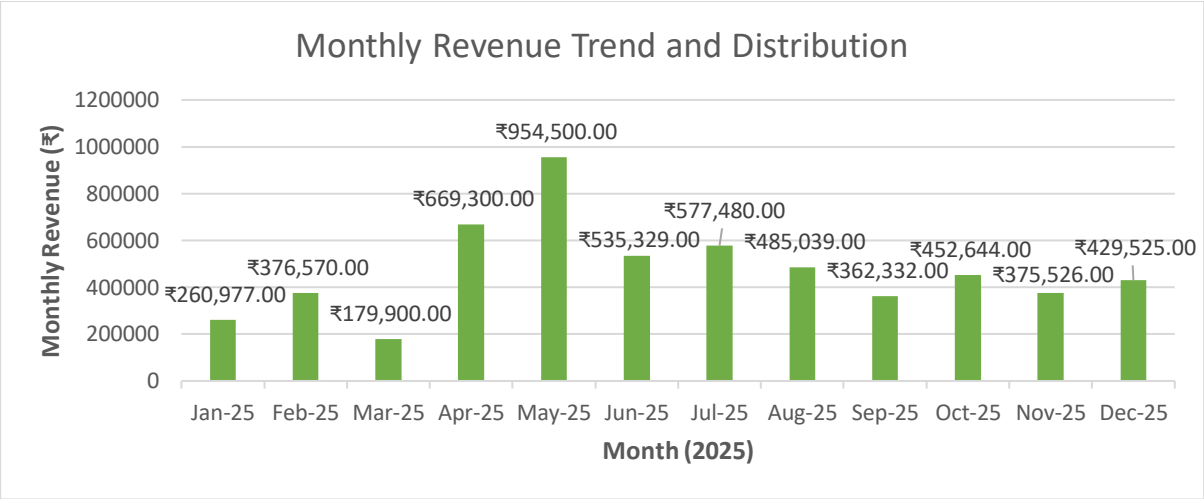

Figure 4: Monthly Revenue Distribution (₹ vs Month)

- **Revenue Peak:**

May recorded the highest revenue based on the occupancy levels. Approximately 20% of annual revenue was generated in just 8–10% of the days, mainly during peak months.

- **Lowest Revenue:**

March had the lowest revenue, which indicated low room occupancy and usage of add-on services. March alone contributed less than 5% of annual revenue, confirming weak revenue extraction during off-peak periods.

- **Revenue Stagnation:**

Although the occupancy was fairly steady following the month of August, the revenue increase was low, indicating that the pricing was not sensitive to demand changes.

- **Insight:**

  Revenue is overly dependent on volume rather than price optimization, exposing the hotel to income instability.

- **Strategic Implications:**

The trend shows the loss of revenue opportunities because it cannot use dynamic pricing when demand is high, and the inability to support pricing when demand is low.

3.5. Relationship Between Occupancy and Revenue

The scatter plot indicates that there is a high positive relationship between the number of rooms occupied and total daily revenue.

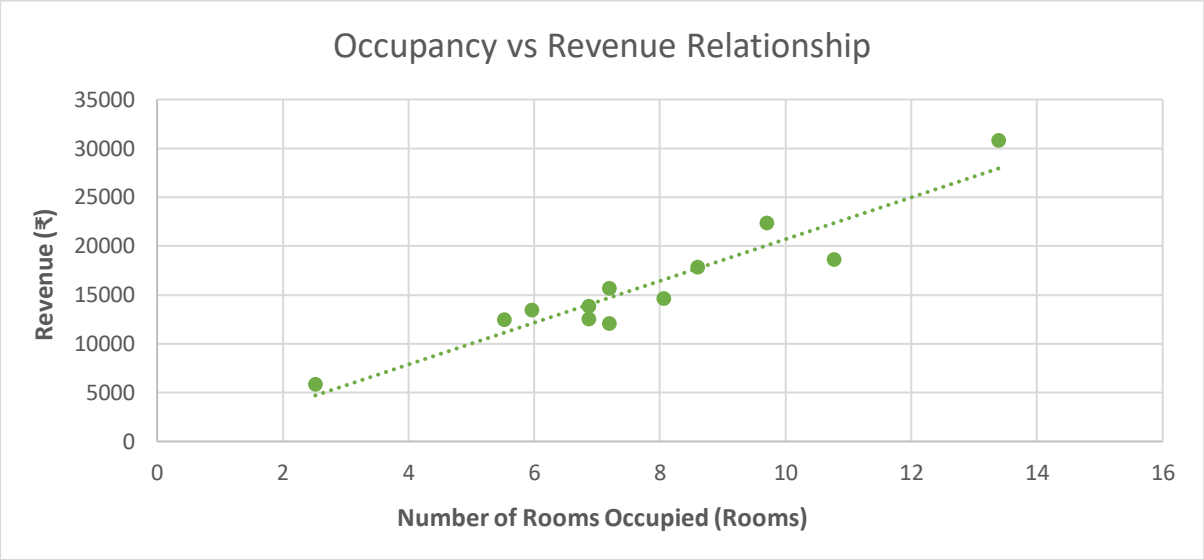

Figure 5: Relationship Between Occupancy and Revenue

- **Direct Relationship:**

There is a positive and strong relationship between the number of rooms occupied and revenue, and this validates that room utilization is the best source of income.

- **Minor Deviations:**

There are minimal differences in revenue, which are affected by the add-on services like additional beds and the sale of cold drinks. Additional services add a daily revenue of about ₹1,800 - ₹2,200 without the add-ons. On high-demand days, add-ons increase overall revenue per booking by 12-18%.

- **Strategic Implications:**

Increasing profitability by occupancy optimization with improved forecasting, promotions, and demand-based pricing does not increase operational costs, which may be improved dramatically.

3.6. Contribution of Add-on Services (Extra Beds and Cold Drinks)

The add-on service analysis indicates that there is a high reliance on peak times of occupancy.

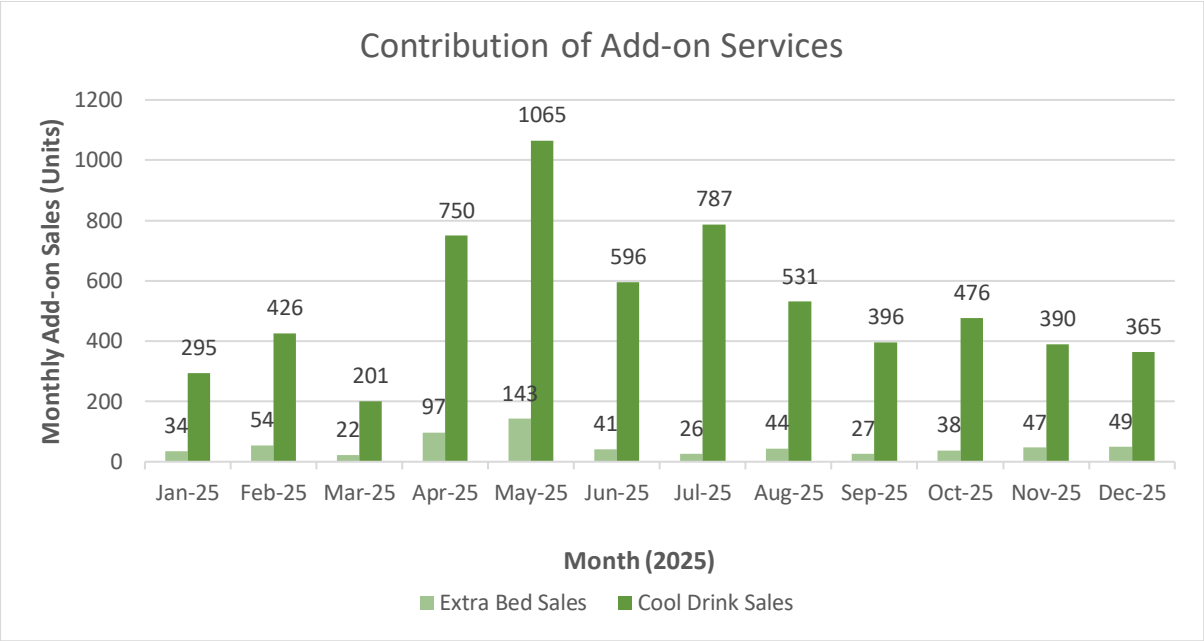

Figure 6: Contribution of Add-on Services (Extra Beds and Cold Drinks)

(This section analyses the relationship between the number of rooms occupied and the total revenue)

- **Add-on services are adding about 10-15 per cent of the total monthly revenue in peak months and less than 5 per cent in off-peak months.**

- **Huge Contribution in Peak months:**

The use of add-ons was a significant phenomenon in high-occupancy months like May and July, and it had a significant impact on the overall revenue.

- **Mismatch during Low-Demand Periods:**

Add-on sales decreased drastically during off-peak months because of a decline of guest's inflow and the absence of active promotion.

- **Insight:**

Add-ons are not maximized when there is low occupancy because of the absence of bundling and specific offers.

- **Strategic Implications:**

Planned upselling programs and package deals on low-occupancy days can be used to transform the slow days into moderate ones.

3.7. Measured Impact of Changing the Strategy of Pricing

In estimating the effect of dynamic pricing, a scenario of a demand-responsive pricing that is conservative was simulated in terms of the past occupancy data.

Assumptions:

- Room rate increased by 10 per cent in the high season times (≈90 days/year)
- Weekday discount of 10-15% when the demand is low (≈180 days/year)
- Average room rate (ARR): ₹2,000
- No increment in fixed operating costs.

Expected Revenue Impact:

1. Peak Period Revenue Gain:
   - Mean occupancy rate: 11 rooms/day
   - Incremental revenue per room: ₹200
   - Estimated Gain:

     11 X ₹ 200 X 90 ≈ ₹1.98 lakh annually
2. Occupancy Stimulation during Off-Peak:
   - Conservative increment of occupancy: +2 rooms/day.
   - Discounted rate: ₹1,700 per room
   - Estimated Gain:

     2 X ₹ 1700 X 180 ≈ ₹6.12 lakh annually
3. Add On Revenue Uplift:
   - Estimated add-on increment: ₹400/day in the event of enhanced occupancy.
   - Annual Gain:

     ₹ 400 X 180 ≈ ₹ 0.72 lakh

(If Occupancy = 80%, the Adjusted Price becomes ₹2000 X (1 + 0.8) = ₹3600. This captures consumer surplus during peak wedding seasons.)

Total Expected Revenue Improvement:

Component	Estimated Annual Gain
Peak price optimization	₹1.98 lakh
Off-peak occupancy increase	₹6.12 lakh
Add-on revenue uplift	₹0.72 lakh
Total Expected Gain	₹8.82 lakh (~18–22%)

Strategic Conclusion:

The dynamic pricing model is viable to raise the annual revenue by ₹8-9 lakh, which is an estimated 18-22 per cent improvement of the current performance without adding to the operational costs. The approach will also minimize volatility of revenue, increase the occupancy of weekdays, and enhance predictability of cash flow, which is the reason why it is a high-impact, low-risk intervention for Hotel SR Paark.

3.8. Overall Findings

It is observed that seasonality is high, the capacity utilization is low, and the Hotel SR Paark faces a risk of concentration on revenue. The hotel is not performing anywhere near the potential of its revenue due to average occupancy of less than 30, a price set in stone and not leveraged on add-on services. Occupancy forecasting, dynamic pricing, and controlled upselling would be a realistic way to increase utilization by 10-15 percentage points and even stabilize weekday and off-peak month revenue, resulting in greater profitability and operational strength.

4. Interpretation of Results

- **Average Occupancy and Capacity Utilization:**
  - The 365 daily records analysis indicates that the average daily occupancy of the hotel is about 6 rooms, which equates to 33.9 percent utilization of the 23 rooms.
  - This indicates that, on an average day, nearly **66.1% of room inventory remains unoccupied**, reflecting substantial underutilization of fixed capacity.
- **Distribution of Occupancy Levels:**
  - The occupancy values are not evenly distributed throughout the year. The number of days in the range 3-7 rooms occupied is more frequent, whereas those with more than 10 rooms occupied are rather isolated.
  - This implies that the hotel's typical operating state is low-to-moderate occupancy, and the overall average occupancy is influenced upward by a limited number of high-occupancy days, masking the frequency of low-demand conditions.
  - Such a one-sided distribution proves that the high occupancy is not year-round but only in several periods.

- **Seasonal Variability in Occupancy:**
  - Aggregation of occupancy data on monthly basis indicates high seasonality.
  - The lowest average of the month is recorded during the first months of the year, and the lowest occupancy rates are recorded in March.
  - Conversely, there is a sharp increase in occupancy in April and May, with the average occupancy enjoying growth over the off-peak months.
  - This trend implies that the demand is very seasonal as opposed to stable.
- **Revenue Variability and Dispersion:**
  - The daily revenue is dispersed, with the minimum and the maximum being ₹2,300 and ₹29,000, respectively.
  - This volatility of revenue is high based on the large range of revenue as compared to the mean revenue.
  - The standard deviation of revenue is quite high and shows that there are large changes daily, with major changes being a result of fluctuating room occupancy.
- **Occupancy–Revenue Relationship:**
  - Analysis of the correlation between the number of rooms occupied and total revenue per day demonstrates a strong positive relation; this means that the amount of revenue is nearly proportional to the number of rooms occupied.
  - The revenue is always greater on days where the room has been fully utilized, which proves that the bookings of the rooms are the primary source of revenue.
- **Behavior of revenue at comparable occupancy rates:**
  - Revenue varies moderately amongst the days, even at the same occupancy levels.
  - This variance can be explained by the fact that there are differences in the usage of add-on services and length of stay that indicate that, though occupancy is the major factor, the impacts of secondary factors on total revenue do exist
- **Add-on Services Contribution Pattern:**
  - The add-on services analysis indicates that the use of add-on services is directly related to occupancy levels.
  - More beds and cold drink sales are made on high occupancy days, and there is little or no add-on revenue on low occupancy days.
  - This proves that add-on revenue is not an independently generated revenue, but an occupancy-based revenue.

- **Lack of Revenue Smoothing:**
  - The results indicate that there are no signs of revenue smoothing between months.
  - Trends in revenue highs and lows are very much similar to occupancy, and this means that there are no compensation strategies like variable pricing or systematic promotions in low demand periods.
- **Time-specific revenue concentration:**
  - A comparatively low percentage of days contributes to a relatively disproportionate amount of revenue.
  - The proportion of annual revenue that is attributed to peak-occupancy months is very high in contrast to off-peak months, which adds a risk to income concentration.
- **Predictability of Demand:**
  - The repetitive nature of the seasonal trend is seen throughout the months; this suggests that the demand is not random.
  - This predictability offers a statistical foundation for making the future occupancy forecasts via time-series techniques.
  - This stability improves the reliability of forecasting models, as historical patterns can be used to estimate future occupancy with lower forecasting error compared to irregular or random demand patterns.
- **Limitations of the Data observed:**
  - The data exhibit a minimal change in the cost in comparison to occupancy rates, indicating that the room rates were not much different in the presence of different demand conditions.
  - Consequently, the fluctuations in revenue are volume-impaired as opposed to price movements.

5. Recommendations to the Business

- **Adopt Weekday Discount Pricing:**
  - To reduce the current low occupancy on weekday nights, the hotel ought to implement discounted room rates of between 10-15% on weekdays when the hotel is not above 50%.
  - This type of targeted pricing approach can be used to trigger the demand at low-utilization seasons without undermining revenue in peak seasons.
  - The effects of such an initiative would be assessed after three months based on occupancy and revenue reports that are supposed to be generated using MS Excel.
- **Embrace Occupancy-Based Price Adjustment:**
  - Hotel SR Paark is advised to adopt the demand-responsive pricing approach whereby the tariff charged on each room is raised (10-20 per cent) during high demand months, especially during the peak time of summer seasons.
  - It is also possible to keep the base pricing in the off-peak months to be competitive.
  - Through historical occupancy data, this strategy can be progressively adopted within a period of six months to enhance the revenue capture during peak demand.
- **Implement the Add-on Service Bundling:**
  - These bundled packages, like Room + Extra Bed or Room + Cold Drinks, should be advertised during the check-in period.
  - Such value-added packages can improve the convenience of the guests as well as generate extra per-room income.
  - Within two months, the add-on revenue can be increased by a percentage of 20 by promoting it by front-desk personnel regularly.
- **Allow Flexible Walk-in Pricing:**
  - In the case of walk-in customers on low occupancy days, price negotiation can be controlled, and discounts should not exceed 15%.
  - Such flexibility would assist in transforming possible walk-ins into a confirmed booking and in securing revenue during low demand time.

- **Enhance Daily Occupancy and Revenue Monitoring:**
    - The real-time monitoring of performance will be achieved by keeping a daily Excel-based tracking system of room occupancy and revenue.
    - Analysis of such data after every working day helps to make prompt corrective measures, including the use of discounts or promotions when the demand is low.
- **Actively Marketing Add-on Services:**
    - The reception staff ought to be conditioned to be aggressive in offering the add-on services to the guests.
    - Placing the add-on price lists clearly in the front desk can also be used to promote sales.
    - The effectiveness of the given initiative will be evaluated with the help of the weekly monitoring of the add-on use.
- **Establish a Customer Loyalty Program:**
    - The hotel can be able to give incentives to its customers who repeat their stay to make the stay more memorable, which would be a 5 percent off or a free cold drink.
    - Repeat visit by outlining booking records or phone number will assist in the creation of customer loyalty and persuade them to return.
- **Introduce Weekday Banquet Pricing Strategy:**
    - Lower prices on banquets on weekdays will be able to lure small meetings, corporate meetings and local events.
    - This would be a way of putting into use the unutilized banquet facilities and should also seek to boost the number of weekday banquet bookings by at least 20.
- **Standardize Digital Record Keeping:**
    - The booking, occupancy, and revenue information must all be stored in the form of MS Excel and default formatted in the standardized form.
    - It will enhance the accuracy of data, ease its analysis and help in the future to forecast and make decisions.

- **Carry out Performance Reviews regularly:**
  - Occupancy and revenue patterns should be reviewed monthly to measure performance as compared to the months before.
  - According to the observed trends, it is possible to vary the pricing strategy, promotional activity, and the service provided to guarantee the ongoing enhancement and maintenance of the revenue.

SUMMARY TABLE:

Identified Problem	Key Evidence from Analysis	Recommendation
Low weekday occupancy	Avg occupancy < 35%, Jan–Mar lowest	Weekday discount pricing (10–15%)
Revenue concentration	20% revenue from <10% days	Dynamic pricing during peak demand
Add-on dependency on occupancy	r = 0.94 between occupancy & add-ons	Add-on bundling & proactive upselling
Underutilized banquet facilities	Low weekday utilization	Weekday banquet pricing strategy
Lack of data-driven decisions	Manual record keeping	Standardized digital tracking

6. Links

- **Dataset & Analysis sheets:**

  https://docs.google.com/spreadsheets/d/1t-R_PTthMI34Qu_5c2TSvYItfKetYL5h/edit?usp=drive_link&ouid=111381051408798734177&rtpof=true&sd=true

- **Proof of Originality:**

  https://drive.google.com/drive/folders/1IQHe6ng5p2Nlmy1XR6S2Ti5xj--L7UpH?usp=drive_link